under *Media* in the palette. Feel free to drag and drop the same into *screen1*. You will see that this gets shown under the screen under 'Non visible components'. To get the time of the day, we would also need to use a clock. So let us drag and drop the clock component from 'Sensors' in the palette onto the screen. This also gets added to the 'Non visible components'. The *screen1* must now look like what's shown in Figure 5.

We have set up the 'Designer view' with elements that we wanted in our screen. We will now go to the 'Blocks view' in the next section and ensure we get the required behaviour to complete our app.

The 'Blocks view'

To get into the 'Blocks view' for our project, click on the 'Blocks' button as shown in *Item 2* in Figure 3. The 'Blocks view' basically allows you to code for events of interest for your app. Here are some generic concepts about 'Blocks':

- The AI 2 interface basically has a philosophy of coding every single item through blocks which could be independent or interconnected (more on this later).
- A block could be a self-sufficient item by itself or it could be a value, a variable or a segment that supplies a result or a procedure.
- Blocks that have a blue box at the top left are configurable and allow you to change their structure.

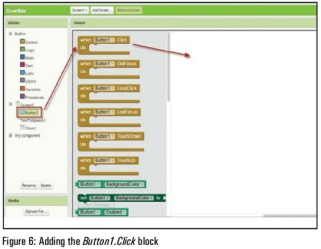

Figure 6: Adding the *Button1.Click* block

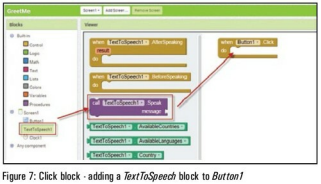

Figure 7: Click block - adding a *TextToSpeech* block to *Button1*

With that brief introduction, let us get into our first block, which is a handler for click events for our *GreetMe* button. To do the same, you have to select *Button1* from *screen1* under the *Blocks* pane, and then click on the 'Button1.Click' block from under the viewer pane as shown in Figure 6.

Remember the first point I made earlier about everything being a block? Now, to make the button click and give a simple greeting, you need to add a *TextToSpeech.Speak* block to the button click handler as shown below. Do this by dragging the same into the *Button.click* block.

Now we need to give something for *TextToSpeech* to speak or say. Let us make this 'Congratulations on your first Android app' in line with the TalkToMe tutorial app from AI 2. You can do the same by dragging a simple text box from under 'Built-in' as shown in Figure 8.

Now type the message into the text box. Once done, your *blocks view* should look like what's shown in Figure 9.

Let us take a brief break here to do some testing of the work done so far. The most important advantage of such an environment is that it is very easy to continually test the application developed so far. Testing can be done using the AI 2 supplied emulator environment or by connecting to an Android smartphone. Let us start with the AI 2 supplied emulator first.

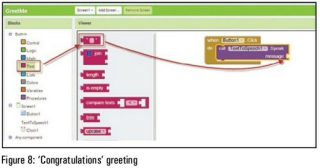

Figure 8: 'Congratulations' greeting

Figure 9: Blocks view with the 'Congratulations' message

Test the app using an emulator

To test using an emulator, the emulator has to be installed on the computer in which you are developing the app. Once you have set it up on your computer, you should be able to connect to the same as shown in Figure 10.

Now you should see an emulator running on your system with the Android icon as shown in Figure 11.

If all goes well, you should be able to see a screen as shown in Figure 12, and be able to hear your congratulatory message on pressing the 'GreetMe' button.